Retail E-Commerce Core

Project Proposal Report & ER Diagram

[Description] [10]

Retail E-Commerce Core is a simple back-office system for a retailer to manage the essentials of an online store. It helps you list products, group them into clear parent categories and categories, keep track of available stock, and record incoming orders with their totals. The aim is to make day-to-day tasks, like updating a price, checking stock before packing, or seeing what was ordered quick and straightforward.

The system keeps customer details together with their delivery information in one place, stores products with their price and live stock levels (on-hand vs reserved), and organizes the catalog under Parent Category → Category. When an order is created, it's saved with its items and bill summary, linked to the customer details used at checkout. Customers can also leave reviews that appear against products so the retailer can monitor feedback. This small, clear structure focuses on what a retailer needs first, managing catalog, track stock, and process orders.

[Problem Statement and Motivation] (1 - 2 paragraphs) **[20]**

Small retailers have their catalog, stock, and orders scattered across various places. This causes overselling, price mismatches, repeated chasing for correct delivery details, and slow packing because order info isn't in one place. There's no reliable history of what was sold, to whom, and with which quantities, making restocking and basic reporting difficult.

We propose a simple system that keeps everything in one database. Products organized under parent categories → categories, each product's current stock (on-hand and reserved), and every order with its items and totals, tied to the exact customer and address used. Basic reviews are stored to capture product feedback. This system reduces mistakes, speeds up fulfillment, and makes day-to-day work easier. Overall, this leads to fewer stock errors, faster order lookup/packing, and simple reports the retailer can trust.

[Key objectives of your system and how you plan to achieve them] (bullet points) **[30]**

1. **Single efficient system**: keep products, categories, customers and addresses, orders, and reviews in one consistent place.

   - All core records live together: Customers**,** Addresses, Parent Categories → Categories, Products (with stock & price), Orders, Order Items, Reviews.

- Each record links to the others (e.g., an order points to the exact customer, address and the exact products), so staff don't jump between files.

2. **Prevent overselling**: track stock_on_hand and stock_reserved so quantities are accurate before packing.

- Every product tracks **stock_on_hand** (what's in store) and **stock_reserved**.

- When an order is created, the system increases *reserved* and decreases stock_on_hand by the amount of order, when delivered, it decreases *reserved*; when cancelled, it decreases *reserved* and adds the amount in stock_on_hand.

3. **Faster order handling**: make it easy to create an order, see its items and totals, and find it later.

- One-click customer details**:** pick the customer and delivery address from their **saved list**, no retyping.

- Easy to find later**:** every order gets a unique number and a status (e.g., placed, shipped, delivered), so you can search and filter quickly.

4. **Simple, reliable reporting**: help restocking and pricing decisions with clear summaries (low stock, orders by date).

- Ready summaries: Low Stock, Orders by Date/Customer, Reviews of Products.

- Because data lives in one place and is linked, these lists are accurate and consistent.

5. **Capture product feedback**: store reviews to spot issues and improve catalog choices.

6. **Data accuracy by design:** use clear IDs and links so products, customers, and orders never get mixed up.

- Every table uses clear IDs; links ensure an order can't reference the wrong product or customer.

- Category selection is controlled via Parent Category → Category to avoid mislabeling.

7. **Keep history correct:** orders store copies of price at the time of sale, so past records don't change after edits.

[Outline the system's main features, operations, and outcomes] (choose top 3-5) **[20]**

1) Catalog Management (Parent Category → Category → Product)

- **Operations**: add/edit parent categories and categories; add products with name, brand, price; assign a category.
- **Outcomes**: tidy product list, fewer listing mistakes, faster product lookup.

2) Stock Control (inside Products)

- **Operations:** update **stock_on_hand** when new stock arrives or new order is placed, or an existing order is cancelled; system adjusts **stock_reserved** when orders are created/cancelled/delivered.

- **Outcomes:** no overselling, clear view of what's available, early low-stock warnings.

3) Order Management (Orders + Order Items)

- **Operations:** create an order, pick customer and address, system auto-calculates subtotal/tax/shipping/grand total; set order status (placed → delivered/cancelled).

- **Outcomes:** faster order entry, consistent bills, easy tracking by order number & status.

4) Customers & Addresses (separate tables)

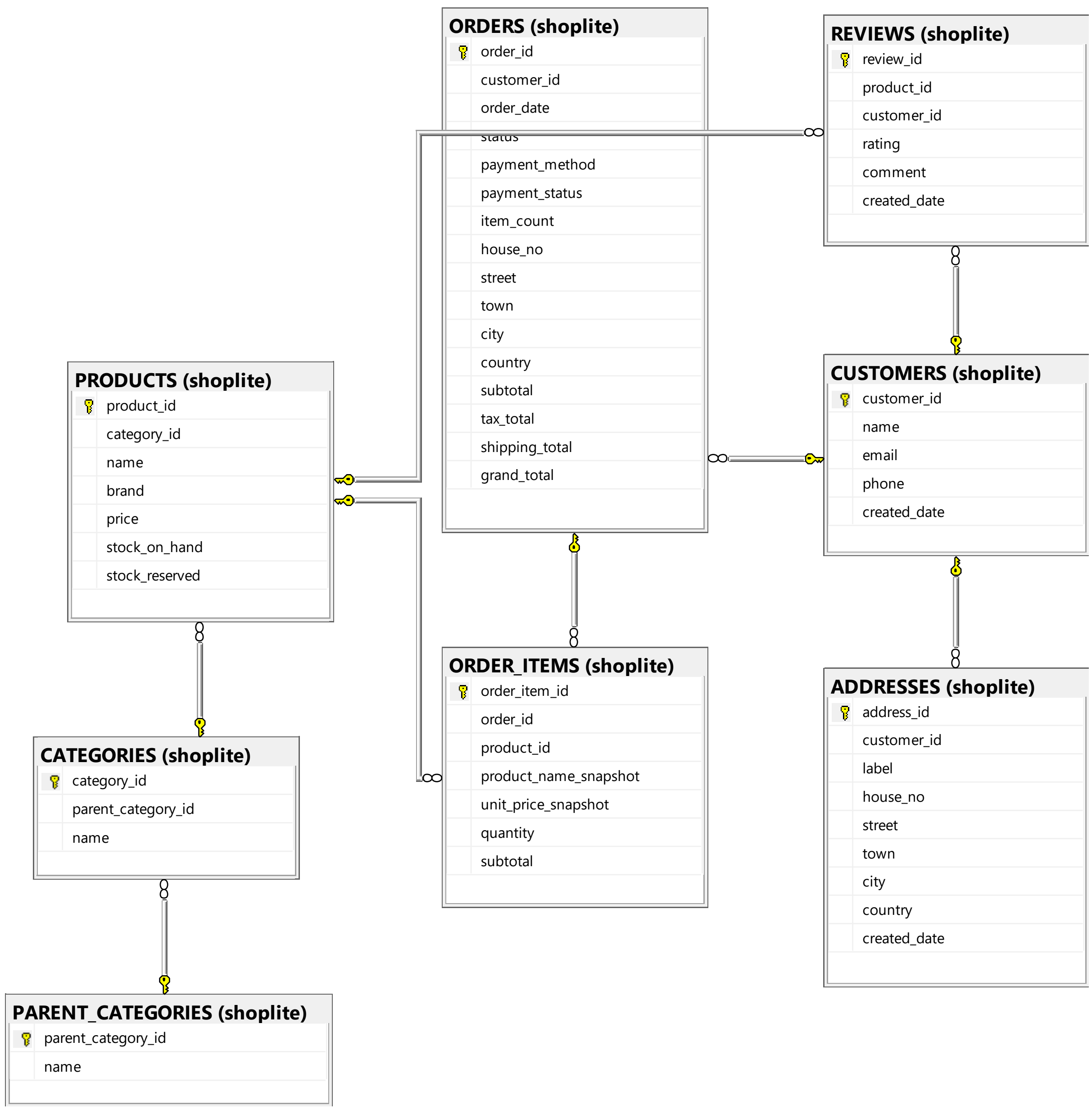
- **Operations:** You add or edit a customer in the Customers table (name, email, phone). You then create one or more entries in the Addresses table for that same customer using customer_id (house no, street, town, city and country). When creating an order, you first select the customer and then choose one of their saved addresses.
- **Outcomes:** Records stay clean because one customer can have many addresses without mixing them up. Order entry is quicker since you pick from saved details, and delivery mistakes drop because the address is linked to the right customer.

5) Product Reviews & Quick Reports

- **Operations:** You record a review as a simple rating (1–5) and short comment linked to both product_id and customer_id. You can also run quick reports: Low Stock (from products' stock_on_hand vs stock_reserved), Orders by Date/Customer (from orders), and Average Rating per Product (from reviews).

- **Outcomes:** These reports show which products are running low, and highlight product issues early through ratings and comments, so you can adjust purchasing or pricing with confidence.

[ER diagram] (just paste your ER diagram picture here) **[20]**



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